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ECE 3364

Circuits and Systems

Lecture Set #1

Complex Power

Version 3

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Overview of this Lecture Set

Complex Power

In this set of lecture notes, we will review material that was previously covered, in ECE 2202. We will review the following topics:

- Sinusoidal Sources and Phasors
- Power with Sinusoids
- Real, Reactive, and Complex Power
- Notation and Units



Textbook Coverage

This material is introduced in different ways in different textbooks. Approximately this same material is covered in your textbook in the following sections:

- Electric Circuits 12th Edition by Nilsson and Riedel: Sections 10.1 through 10.6

Power in the Sinusoidal Steady State (Complex Power)

We studied Phasor Transforms. Using these transforms, we can find things we want to know, more quickly and more easily.

Now we are going to do a similar thing with power absorbed or delivered in circuits with sinusoidal sources. Again, we will only consider what is happening in the steady state. We will find that:

- The use of phasors and transforms can be used for power calculations, and
- Some very useful new concepts will help us deliver more power to where we want it, with fewer losses at the same time.

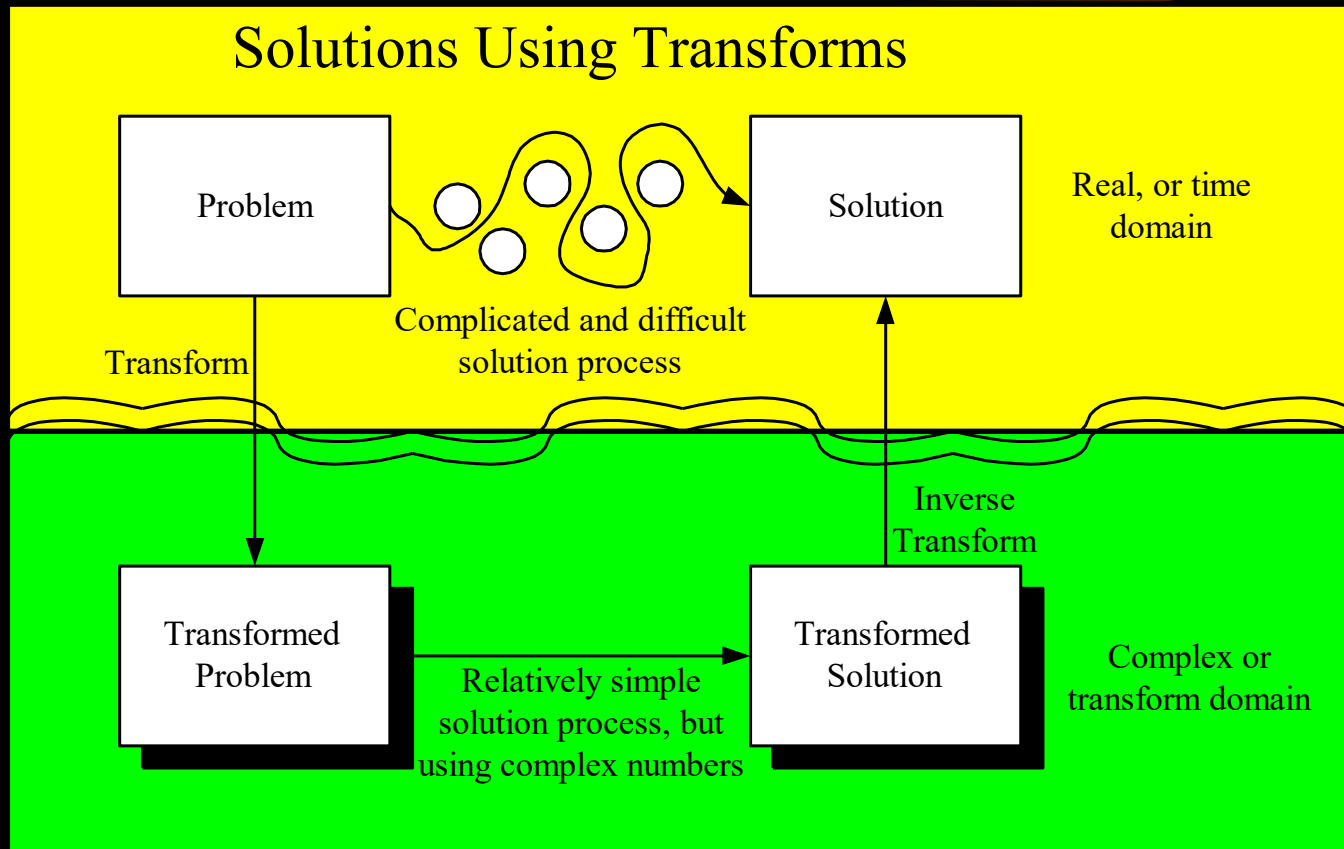


The power we use is often sinusoidal. That is, the wall plugs, and some other sources, are voltages and currents that vary as a sine wave. Thus, this subject is very useful to us.



AC Circuit Analysis Using Transforms

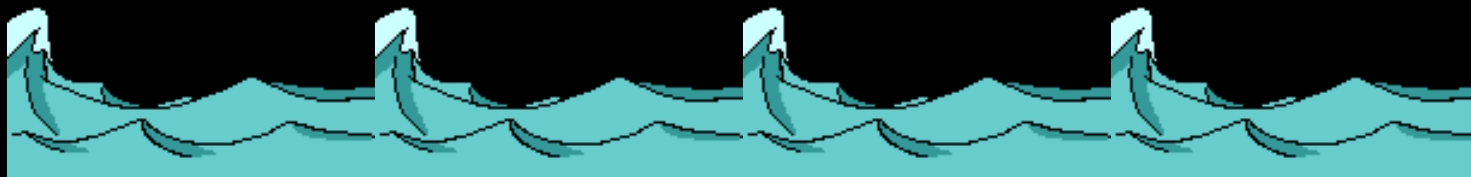
Let us remember first and foremost that the end goal is to find the solution to real problems. We will use the transform domain, and discuss quantities which are complex, but obtaining the real solution is the goal.





Power with Sinusoidal Voltages and Currents

- It is important to remember that nothing has really changed with respect to the power expressions that we are looking for. Power is still obtained by multiplying voltage and current.
- The fact that the voltage and current are sine waves or cosine waves does not change this formula.





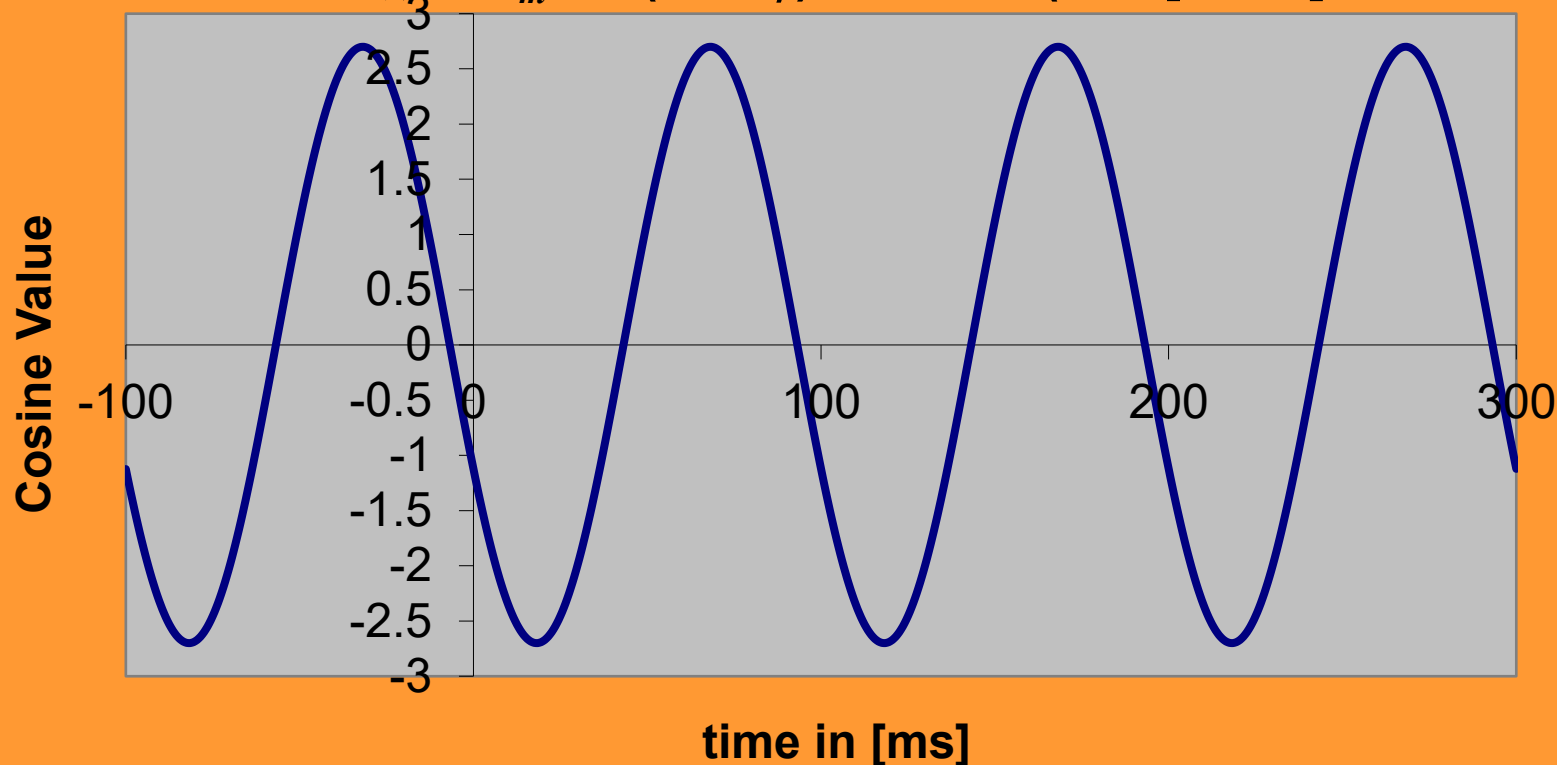
Some Review – Sinusoids

The figure below is a cosine function. Note that in this equation there are three parameters, the amplitude (X_m), the frequency (ω), and the phase (ϕ). The time, t , is the independent variable. The function is

$$x(t) = X_m \cos(\omega t + \phi).$$

Cosine with Phase Shift

$$x(t) = X_m \cos(\omega t + \phi) = 2.7 \cos(62.8[\text{rad/s}]t + 114.6^\circ)$$





Definition of RMS – Review

We are now going to review an important term, the **rms value** of a voltage or current. This was covered before. The rms value, also called the **effective value**, has the most meaning in terms of power calculations.

It is so useful, that we will redefine our phasor transforms in terms of rms values, to make our formulas simpler and easier to use. Thus, it is worth the time to review rms concepts.





Derivation of RMS – 1

We want the effective value that could be used in power calculations, for average power, in the formula

$$P_{ave} = V_{rms} I_{rms}.$$

We will do the derivation for a resistance, since we want the formula to work with the resistance power formulas. Let's arbitrarily choose to work with the voltage. What we want to get is a value that will work in the formula,

$$P_{ave} = \frac{(V_{rms})^2}{R}.$$

With T as the period, the average value of the power is obtained by the formula,

$$P_{ave} = \frac{1}{T} \int_{t_0}^{t_0+T} \left(\frac{v^2(t)}{R} \right) dt.$$





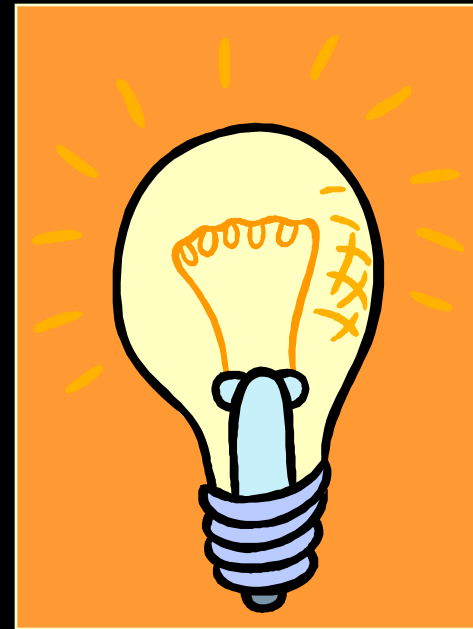
Derivation of RMS – 2

Now, to get the formula, we simply set the two equations from the previous slide equal to each other,

$$P_{ave} = \frac{(V_{rms})^2}{R} = \frac{1}{T} \int_{t_0}^{t_0+T} \left(\frac{v^2(t)}{R} \right) dt.$$

Now, we need to simplify. The resistance is assumed to be a constant, and so it can be taken out of the integral. When we multiply both sides by R , we get

$$(V_{rms})^2 = \frac{1}{T} \int_{t_0}^{t_0+T} (v^2(t)) dt.$$





Derivation of RMS – 3

Finally, we can solve for the rms value of the voltage, by taking the square root of both sides,

$$V_{rms} = \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} (v^2(t)) dt}.$$

This is the result that we have been working toward. We only need to interpret this result. We have taken the periodic voltage, $v(t)$, and squared it. Then, by integrating it over a period and dividing by the period, we are taking the mean value of the squared function. Finally, we take the square root of the mean value of the squared function. We call this **rms**.





RMS Value of a Sinusoid

The rms value for a general periodic function, $x(t)$, is

$$X_{rms} = \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} (x^2(t)) dt}.$$

Now, this was derived for any periodic function. The function must be periodic for the formula for the mean value to apply.

If we perform the calculus to get the rms value for a sinusoid, we find the rms value is equal to the zero-to-peak value (or amplitude) divided by the square root of 2, or

$$X_{rms} = \frac{X_m}{\sqrt{2}}.$$

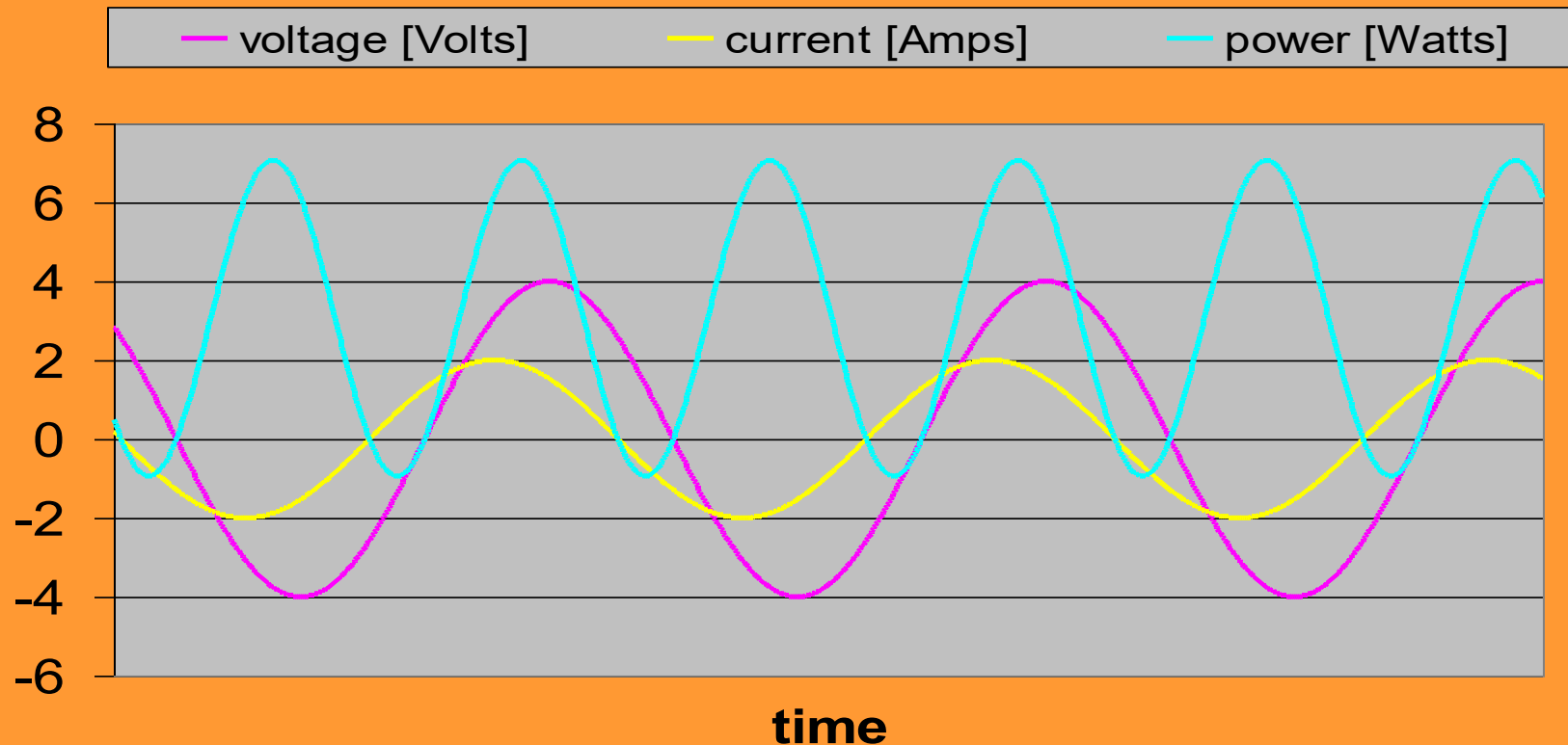
Remember, this only holds
for **sinusoids**!



Power with Sinusoids – 1

If we have sinusoidal voltages and currents, we can get the power by multiplying the two. We could plot the power as shown below. These curves were obtained by taking a couple of arbitrary sinusoids for voltage and current. If you change the magnitudes and phases, these curves will change.

Power with Sinusoids





Power with Sinusoids – 2

Let us approach this same issue using equations. We assume that our voltage and current have the formulas,

$$v(t) = V_m \cos(\omega t + \theta_v), \text{ and}$$
$$i(t) = I_m \cos(\omega t + \theta_i).$$

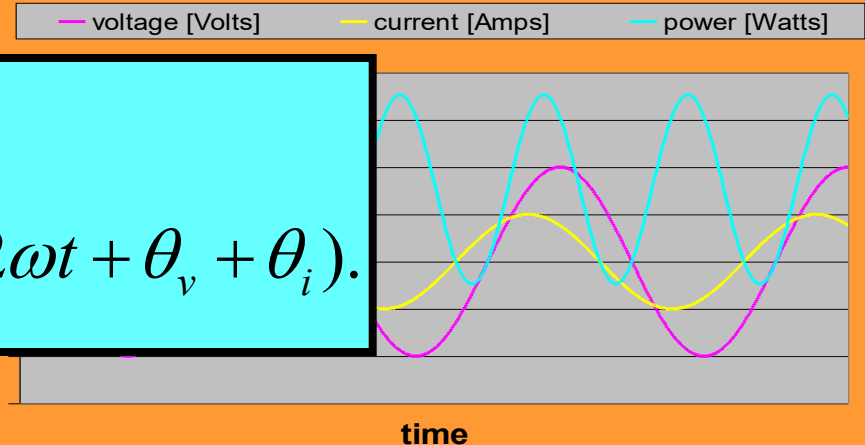
Now, we can use the formula from trigonometry, which most of us learned but forgot,

$$\cos(a) \cos(b) = \frac{1}{2} \cos(a - b) + \frac{1}{2} \cos(a + b).$$

Applying this here, we get

$$p(t) = v(t)i(t) =$$
$$= \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) + \frac{V_m I_m}{2} \cos(2\omega t + \theta_v + \theta_i).$$

Power with Sinusoids



Power with Sinusoids – 3

In the last slide we found that

$$\begin{aligned} p(t) &= v(t)i(t) = \\ &= \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) + \frac{V_m I_m}{2} \cos(2\omega t + \theta_v + \theta_i). \end{aligned}$$

Now, we can use another formula from trigonometry,

$$\cos(a + b) = \cos(a) \cos(b) - \sin(a) \sin(b).$$

Applying this here in
the second term, with

$$\begin{aligned} a &= 2\omega t, \text{ and} \\ b &= \theta_v + \theta_i, \end{aligned}$$

we get

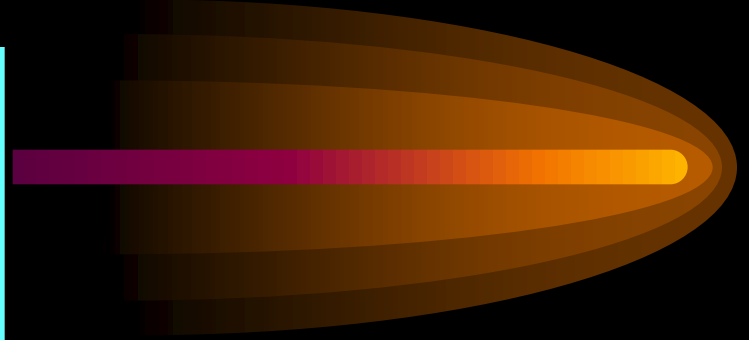
$$\begin{aligned} p(t) &= v(t)i(t) = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) + \\ &+ \frac{V_m I_m}{2} \cos(\theta_v + \theta_i) \cos(2\omega t) + \\ &- \frac{V_m I_m}{2} \sin(\theta_v + \theta_i) \sin(2\omega t). \end{aligned}$$



Power with Sinusoids – 4

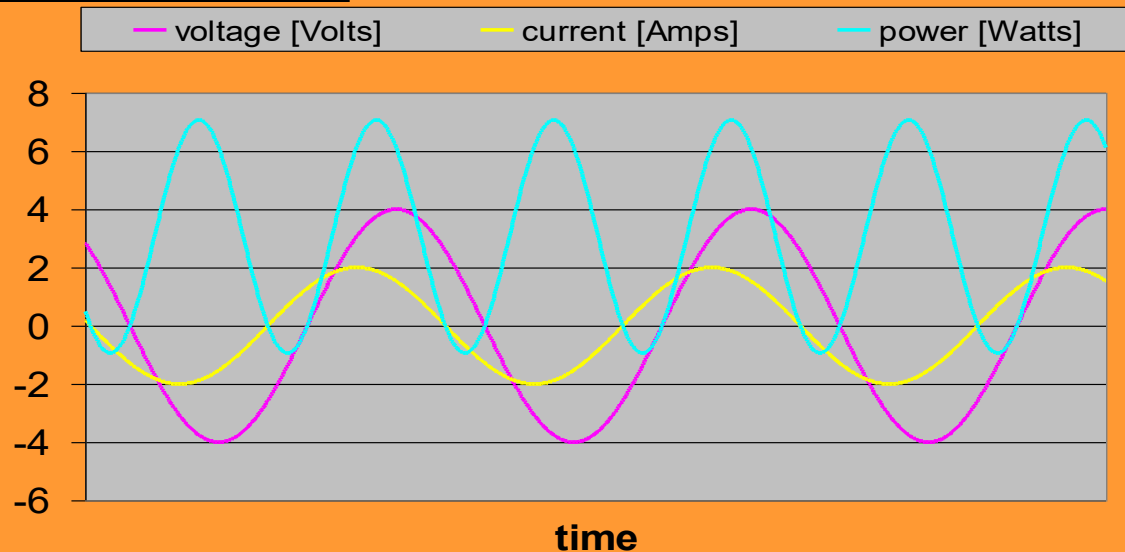
So, we have that

$$p(t) = v(t)i(t) = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) + \frac{V_m I_m}{2} \cos(\theta_v + \theta_i) \cos(2\omega t) - \frac{V_m I_m}{2} \sin(\theta_v + \theta_i) \sin(2\omega t).$$



Now, we can look at the plot that we had, and it can make more sense. Let us note some special properties in the slides that follow.

Power with Sinusoids



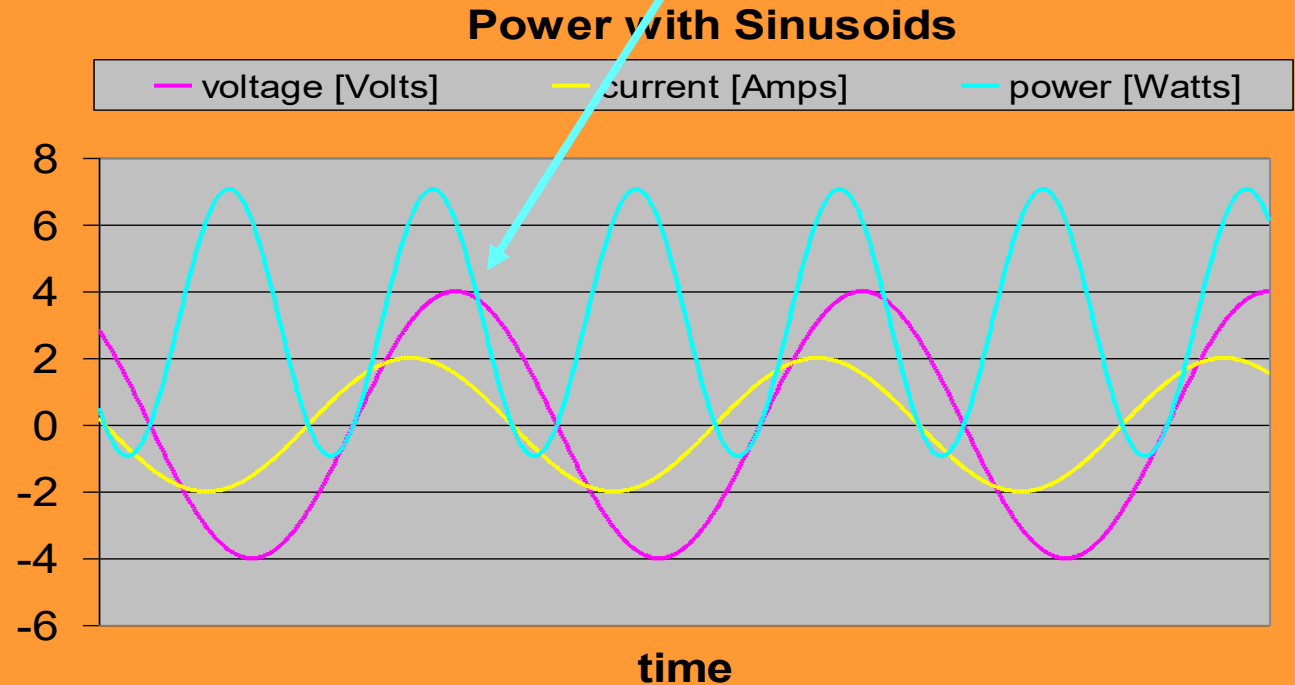


Power with Sinusoids – Note 1

$$p(t) = v(t)i(t) = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) + \frac{V_m I_m}{2} \cos(\theta_v + \theta_i) \cos(2\omega t) - \frac{V_m I_m}{2} \sin(\theta_v + \theta_i) \sin(2\omega t).$$

The power curve is not centered at zero, so the average is not zero.

First, note that even though $v(t)$ and $i(t)$ were zero-mean sinusoids, the product, $p(t)$, does not generally have a zero mean.





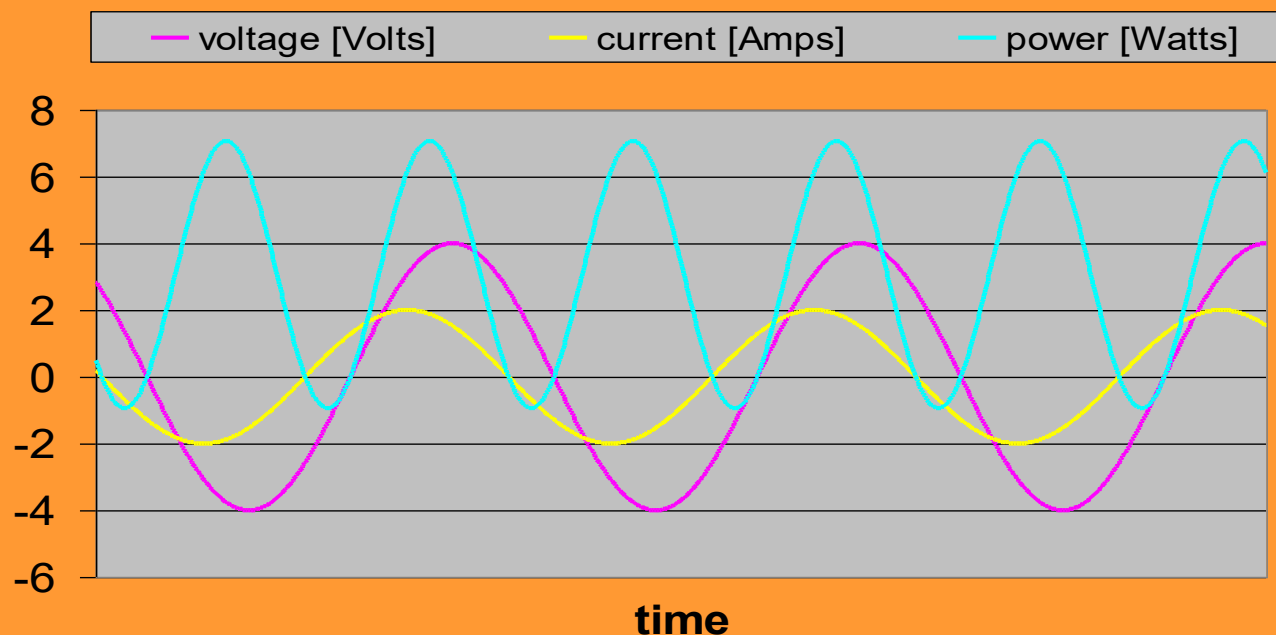
Power with Sinusoids – Note 2

$$p(t) = v(t)i(t) = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) + \frac{V_m I_m}{2} \cos(\theta_v + \theta_i) \cos(2\omega t) - \frac{V_m I_m}{2} \sin(\theta_v + \theta_i) \sin(2\omega t).$$

$$P_{AVERAGE} = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i).$$

In fact, the mean, or average value, of the power is equal to the first term of this equation, since the average of the sinusoids is zero.

Power with Sinusoids





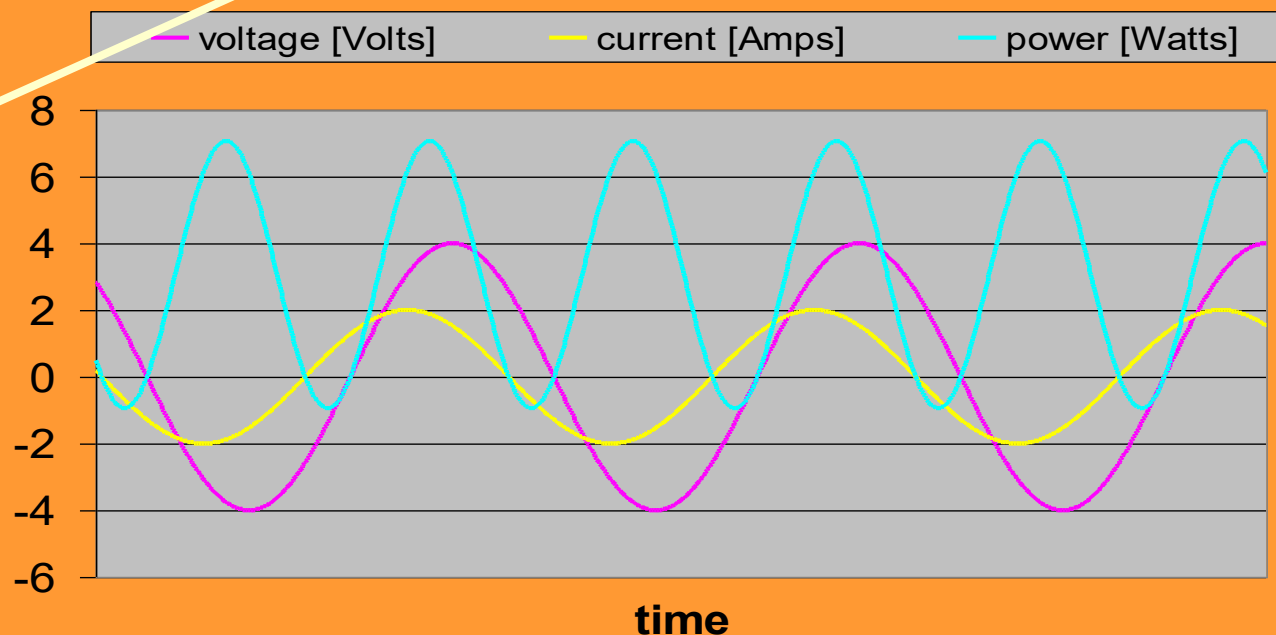
Power with Sinusoids – Note 3

$$p(t) = v(t)i(t) = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) +$$
$$+ \frac{V_m I_m}{2} \cos(\theta_v + \theta_i) \cos(2\omega t) +$$
$$- \frac{V_m I_m}{2} \sin(\theta_v + \theta_i) \sin(2\omega t).$$

$$p_{AVERAGE} = P = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i).$$

We use a capital letter P to represent the average value of the power, $p(t)$. The average power is very useful to know.

Power with Sinusoids





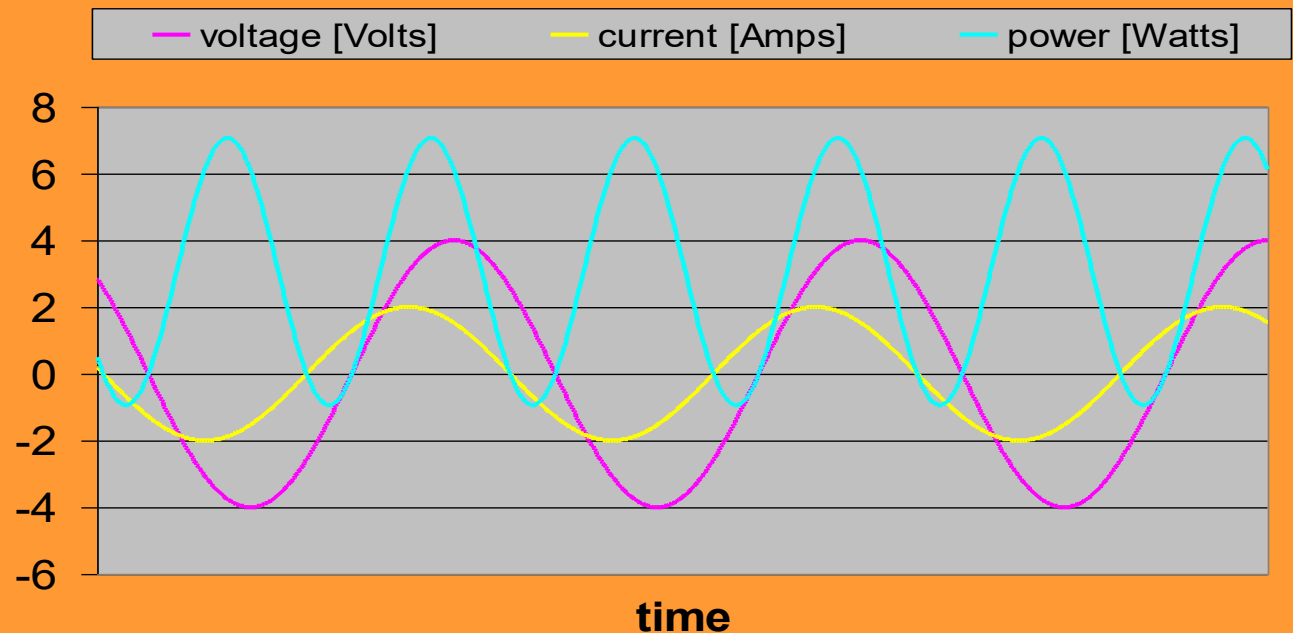
Power with Sinusoids – Note 4

$$p(t) = v(t)i(t) = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) +$$
$$+ \frac{V_m I_m}{2} \cos(\theta_v + \theta_i) \cos(2\omega t) +$$
$$- \frac{V_m I_m}{2} \sin(\theta_v + \theta_i) \sin(2\omega t).$$

$$P_{AVERAGE} = P = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i).$$

This average power P is a function of the magnitudes of the voltage and current, but also of the **difference in phase** between voltage and current.

Power with Sinusoids



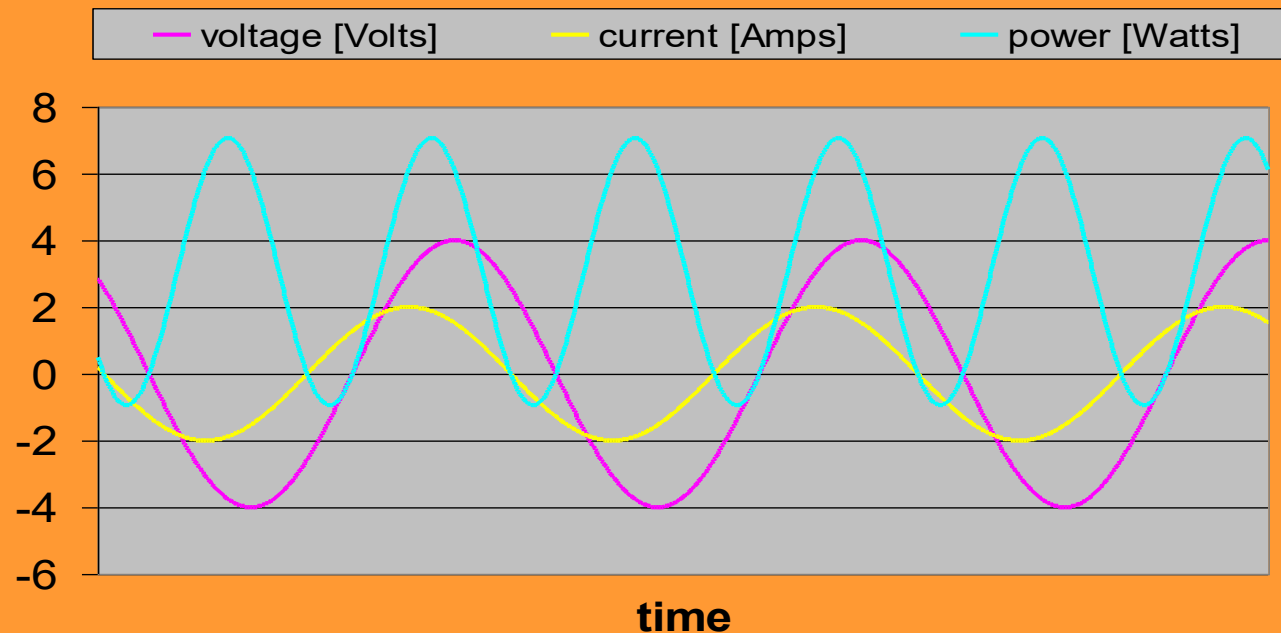
Power with Sinusoids – Note 5

$$\begin{aligned}
 p(t) = v(t)i(t) &= \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) + \\
 &+ \frac{V_m I_m}{2} \cos(\theta_v + \theta_i) \cos(2\omega t) + \\
 &- \frac{V_m I_m}{2} \sin(\theta_v + \theta_i) \sin(2\omega t).
 \end{aligned}$$

$$p_{AVERAGE} = P = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i).$$

The power varies with time, and is in fact sinusoidal with a frequency twice that of the voltage and current.

Power with Sinusoids



Shifting the Time Axis

For notational reasons, electrical engineers like to take the general case, which we have been considering, and then shift the time axis, so that the phase of the current is zero. The new phase of the voltage is now reduced by the phase of the current, and now is $\theta_v - \theta_i$. We redefine this phase of the voltage as θ , and get a new set of formulas, below.

$$v(t) = V_m \cos(\omega t + \theta_v) \text{ and } i(t) = I_m \cos(\omega t + \theta_i)$$

General
case

$$p(t) = v(t)i(t) = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) + \frac{V_m I_m}{2} \cos(\theta_v + \theta_i) \cos(2\omega t) - \frac{V_m I_m}{2} \sin(\theta_v + \theta_i) \sin(2\omega t)$$

$$p_{AVERAGE} = P = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i)$$

$$v(t) = V_m \cos(\omega t' + \theta) \text{ and } i(t) = I_m \cos(\omega t')$$

Shifted
case

$$p(t') = v(t')i(t') = \frac{V_m I_m}{2} \cos(\theta) + \frac{V_m I_m}{2} \cos(\theta) \cos(2\omega t') - \frac{V_m I_m}{2} \sin(\theta) \sin(2\omega t')$$

$$p_{AVERAGE} = P = \frac{V_m I_m}{2} \cos(\theta)$$

Shifting the Time Axis – Note 1

Some of you may be disturbed by the relationship between the two cases below. It may look like we have used $\theta = \theta_v + \theta_i$ in some cases, and $\theta = \theta_v - \theta_i$ in other cases. We have not. Remember that the t in the General case is different from the t' in the Shifted case.

$$v(t) = V_m \cos(\omega t + \theta_v) \text{ and } i(t) = I_m \cos(\omega t + \theta_i)$$

General
case

$$p(t) = v(t)i(t) = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) + \frac{V_m I_m}{2} \cos(\theta_v + \theta_i) \cos(2\omega t) - \frac{V_m I_m}{2} \sin(\theta_v + \theta_i) \sin(2\omega t)$$

$$p_{AVERAGE} = P = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i)$$

$$v(t) = V_m \cos(\omega t' + \theta) \text{ and } i(t) = I_m \cos(\omega t')$$

Shifted
case

$$p(t') = v(t')i(t') = \frac{V_m I_m}{2} \cos(\theta) + \frac{V_m I_m}{2} \cos(\theta) \cos(2\omega t') - \frac{V_m I_m}{2} \sin(\theta) \sin(2\omega t')$$

$$p_{AVERAGE} = P = \frac{V_m I_m}{2} \cos(\theta)$$

Shifting the Time Axis – Note 2

Since we shifted the time axis, we changed the t to a t' in the previous slide. However, the original choice of t was arbitrary, and there is no reason to keep the prime any longer. Therefore, for the rest of this material, we will use the notation below. Remember that θ is the phase of the voltage with respect to the phase of the current. This way of expressing the phases will be useful to us. We will generally get θ by subtracting the phase of the current from the phase of the voltage.

Given that:

$$v(t) = V_m \cos(\omega t + \theta) \text{ and}$$

$$i(t) = I_m \cos(\omega t)$$

$$P_{AVERAGE} = P = \frac{V_m I_m}{2} \cos(\theta)$$

$$\begin{aligned} p(t) = v(t)i(t) &= \frac{V_m I_m}{2} \cos(\theta) + \\ &+ \frac{V_m I_m}{2} \cos(\theta) \cos(2\omega t) + \\ &- \frac{V_m I_m}{2} \sin(\theta) \sin(2\omega t) \end{aligned}$$

Power with Sinusoidal Sources

The formulas below are important, and are the beginning of the concepts that follow in the next two parts. We have found two things. First, the average power is a function of the product of the magnitudes of the voltage and current, and also a function of the difference between the phase of the voltage and current, θ . Second, we found that the expression for the power as a function of time has a constant term, which is that average value, and sinusoidal terms at twice the frequency of the voltage and current.

Given that:

$$v(t) = V_m \cos(\omega t + \theta) \text{ and}$$

$$i(t) = I_m \cos(\omega t)$$

$$P_{AVERAGE} = P = \frac{V_m I_m}{2} \cos(\theta)$$

$$\begin{aligned} p(t) = v(t)i(t) &= \frac{V_m I_m}{2} \cos(\theta) + \\ &+ \frac{V_m I_m}{2} \cos(\theta) \cos(2\omega t) + \\ &- \frac{V_m I_m}{2} \sin(\theta) \sin(2\omega t) \end{aligned}$$



So what is the point of all this?

- This is a good question. First, our premise is that since electric power is usually distributed as sinusoids, the issue of sinusoidal power is important.
- In addition, there are some significant problems that will arise when we connect loads that act like inductors to our power lines. This problems can be addressed using phasor analysis, and some additional concepts that we will lay out in the next slides. These concepts involved quantities called **real and reactive power**.





Real Power and Reactive Power

We have determined the formulas for power when we have sinusoidal voltages and currents.

Now we are going to use these formulas to develop a way of approaching systems with sinusoidal sources that allow us to improve the performance of large electric motors, while at the same time reducing the loss of power in the transmission lines that carry the power from generating stations. We will find that:

- A new quantity called **reactive power** is a measure of the power that will be returned from the load to the source later in the same period of the sinusoid.
- Reducing this reactive power in the load is a good thing.
- Using phasor analysis will make it relatively simple to find this reactive power.



The power lines, which connect us from distant power generating systems, result in lost power. However, this lost power can be reduced by adjustments in the loads. This led to the use of the concept of reactive power.



Power as a Function of Time

We start with the equation for power as a function of time, when the voltage and current are sinusoids. We found that

Given that:

$$v(t) = V_m \cos(\omega t + \theta) \text{ and}$$

$$i(t) = I_m \cos(\omega t); \text{ then}$$

$$p(t) = v(t)i(t) = \frac{V_m I_m}{2} \cos(\theta) + \frac{V_m I_m}{2} \cos(\theta) \cos(2\omega t) + \frac{V_m I_m}{2} \sin(\theta) \sin(2\omega t).$$

The terms set off in red and blue above have meaning and are useful, and so we will give them names.





Definition of Real Power

We define the term in **red** to be the **Real Power**. We use the capital letter P for this. Note that we have already shown that this is the average power as well.



Given that:

$$v(t) = V_m \cos(\omega t + \theta) \text{ and}$$

$$i(t) = I_m \cos(\omega t); \text{ then}$$

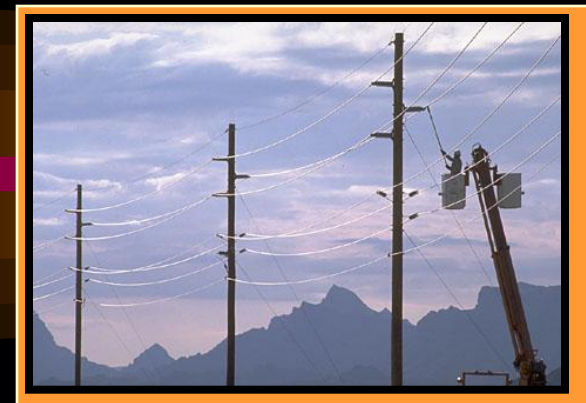
$$\begin{aligned} p(t) = v(t)i(t) &= \frac{V_m I_m}{2} \cos(\theta) + \\ &+ \frac{V_m I_m}{2} \cos(\theta) \cos(2\omega t) + \\ &- \frac{V_m I_m}{2} \sin(\theta) \sin(2\omega t). \end{aligned}$$

$$\text{Real Power} = p_{AVERAGE} = P = \frac{V_m I_m}{2} \cos(\theta).$$



Definition of Reactive Power

We define the term in blue to be the **Reactive Power**. We use the capital letter Q for this. The meaning for this will be explained in more depth later.



Given that:

$$v(t) = V_m \cos(\omega t + \theta) \text{ and}$$

$$i(t) = I_m \cos(\omega t); \text{ then}$$

$$\begin{aligned} p(t) = v(t)i(t) &= \frac{V_m I_m}{2} \cos(\theta) + \\ &+ \frac{V_m I_m}{2} \cos(\theta) \cos(2\omega t) + \\ &- \frac{V_m I_m}{2} \sin(\theta) \sin(2\omega t). \end{aligned}$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$



Meaning of Reactive Power – Part 1

For most students, the meaning of Real Power, P , is fairly clear. Real Power is the average power. The Reactive Power, Q , is much less obvious. To explain it, we will begin by noting that in the phasor domain, we have

$$v(t) = V_m \cos(\omega t + \theta) \text{ and}$$

$$i(t) = I_m \cos(\omega t); \text{ then}$$

$$V(\omega) = V_m \angle \theta \text{ and}$$

$$I(\omega) = I_m. \text{ Thus, the impedance will be}$$

$$Z = \frac{V(\omega)}{I(\omega)} = \frac{V_m}{I_m} \angle \theta.$$

$$\begin{aligned} p(t) = v(t)i(t) &= \frac{V_m I_m}{2} \cos(\theta) + \\ &+ \frac{V_m I_m}{2} \cos(\theta) \cos(2\omega t) + \\ &- \frac{V_m I_m}{2} \sin(\theta) \sin(2\omega t). \end{aligned}$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$

Meaning of Reactive Power – Part 2 (Resistive Case)

Let us look at some special cases. Take the case where our circuit is purely resistive, that is, it could be modeled using only resistors. In this case the impedance is real, which means that θ is equal to zero. We get that

$$v(t) = V_m \cos(\omega t + 0) \text{ and}$$

$$i(t) = I_m \cos(\omega t); \text{ then}$$

$$V(\omega) = V_m \angle 0 \text{ and}$$

$$I(\omega) = I_m. \text{ Thus, the impedance will be}$$

$$Z = \frac{V(\omega)}{I(\omega)} = \frac{V_m}{I_m} \angle 0.$$

$$p(t) = v(t)i(t) = \frac{V_m I_m}{2} + \frac{V_m I_m}{2} \cos(2\omega t).$$

$$\text{Real Power} = P = \frac{V_m I_m}{2}.$$

$$\text{Reactive Power} = Q = 0.$$

In the resistive case, where θ is equal to zero, this reactive power is zero.

Meaning of Reactive Power – Part 3 (Inductive Case)

Let us look at another special case. Take the case where our circuit is purely inductive, and could be modeled using only inductors. In this case the impedance is positive and imaginary, and θ is equal to 90° . We get that

$$v(t) = V_m \cos(\omega t + 90^\circ) \text{ and}$$

$$i(t) = I_m \cos(\omega t); \text{ then}$$

$$V(\omega) = V_m \angle 90^\circ \text{ and}$$

$$I(\omega) = I_m. \text{ Thus, the impedance will be}$$

$$Z = \frac{V(\omega)}{I(\omega)} = \frac{V_m}{I_m} \angle 90^\circ.$$

$$p(t) = v(t)i(t) = -\frac{V_m I_m}{2} \sin(2\omega t).$$

$$\text{Real Power} = P = 0.$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2}.$$

In the inductive case, where θ is equal to 90° , the real power is zero. This should make sense, since with inductors energy is stored in the magnetic field, but later returned to the circuit.

Meaning of Reactive Power – Part 4 (Capacitive Case)

Let us look at a third special case. Take the case where our circuit is purely capacitive, and could be modeled using only capacitors. In this case the impedance is negative and imaginary, and θ is equal to -90° . We get that

$$v(t) = V_m \cos(\omega t - 90^\circ) \text{ and}$$

$$i(t) = I_m \cos(\omega t); \text{ then}$$

$$V(\omega) = V_m \angle -90^\circ \text{ and}$$

$$I(\omega) = I_m. \text{ Thus, the impedance will be}$$

$$Z = \frac{V(\omega)}{I(\omega)} = \frac{V_m}{I_m} \angle -90^\circ.$$

$$p(t) = v(t)i(t) = \frac{V_m I_m}{2} \sin(2\omega t).$$

$$\text{Real Power} = P = 0.$$

$$\text{Reactive Power} = Q = -\frac{V_m I_m}{2}.$$

In the capacitive case, where θ is equal to -90° , the real power is zero. This should make sense, since with capacitors energy is stored in the electric field, but later returned to the circuit.

Meaning of Reactive Power – Part 5 (Conclusion)

So, we have the following situation. The Real Power, P , is the average power, and is the power associated with resistances. The inductors and capacitors take energy in during the first half cycle of a sinusoid, but then give all of that energy back in the second half cycle. The Reactive Power, Q , is used as a measure of the energy that is given to the inductors and capacitors, and then returned later.

$$v(t) = V_m \cos(\omega t + \theta) \text{ and } i(t) = I_m \cos(\omega t); \text{ then } Z = \frac{V(\omega)}{I(\omega)} = \frac{V_m}{I_m} \angle \theta.$$

$$p(t) = v(t)i(t) = \frac{V_m I_m}{2} \cos(\theta) + \frac{V_m I_m}{2} \cos(\theta) \cos(2\omega t) - \frac{V_m I_m}{2} \sin(\theta) \sin(2\omega t).$$

$$\text{Real Power} = p_{AVERAGE} = P = \frac{V_m I_m}{2} \cos(\theta)$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$



Meaning of Reactive Power – (Note)

The Reactive Power, Q , is used as a measure of the energy that is given to the inductors and capacitors, and then returned later. It can be shown that the energy given in the first half cycle is Q/ω . This energy is returned from the inductors and capacitors in the second half cycle. The Reactive Power is important in applications relating to transferring power over transmission lines.

$$v(t) = V_m \cos(\omega t + \theta) \text{ and} \\ i(t) = I_m \cos(\omega t); \text{ then} \\ Z = \frac{V(\omega)}{I(\omega)} = \frac{V_m}{I_m} \angle \theta.$$

$$p(t) = v(t)i(t) = \frac{V_m I_m}{2} \cos(\theta) + \\ + \frac{V_m I_m}{2} \cos(\theta) \cos(2\omega t) + \\ - \frac{V_m I_m}{2} \sin(\theta) \sin(2\omega t).$$

$$\text{Real Power} = p_{AVERAGE} = P = \frac{V_m I_m}{2} \cos(\theta)$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$



Power Terminology - 1

Several quantities are used so often in power calculations that they are given specific names. The first definition, the **Power Factor Angle**, involves the phase angle between voltage sinusoid and the current sinusoid. We have used the symbol θ for this here. This is shown assuming that the phase angle of the current sinusoid, $i(t)$, is zero. Some textbooks use arbitrary phases for current and voltage, and call them θ_i , and θ_v . In this case, the angle of interest would be the angle of the voltage with respect to the angle of the current, or $\theta_v - \theta_i$.

Power Factor Angle $\equiv \theta$.

Using the alternative notation, we would say

Power Factor Angle $\equiv \theta_v - \theta_i$.

This special symbol indicates that we are defining a new quantity.

$$v(t) = V_m \cos(\omega t + \theta) \text{ and}$$

$$i(t) = I_m \cos(\omega t); \text{ then}$$

$$Z = \frac{V(\omega)}{I(\omega)} = \frac{V_m}{I_m} \angle \theta.$$

$$\text{Real Power} = p_{\text{AVERAGE}} = P = \frac{V_m I_m}{2} \cos(\theta)$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$

Power Terminology - 2

Several quantities are used so often in power calculations that they are given specific names. The second definition, the **Power Factor**, is the cosine of the phase angle between voltage sinusoid and the current sinusoid. Again, we define it using both of the possible notations for the phase angles. Remember that the definitions at right are the ones that we will use in these notes, where we assume that the phase of the current, θ_i , has been set to zero.

Power Factor $\equiv pf \equiv \cos \theta$.

Using the alternative notation, we would say

Power Factor $\equiv pf \equiv \cos(\theta_v - \theta_i)$.

Note that pf is used as a common abbreviation for power factor.

$$v(t) = V_m \cos(\omega t + \theta) \text{ and}$$

$$i(t) = I_m \cos(\omega t); \text{ then}$$

$$Z = \frac{V(\omega)}{I(\omega)} = \frac{V_m}{I_m} \angle \theta.$$

$$\text{Real Power} = p_{AVERAGE} = P = \frac{V_m I_m}{2} \cos(\theta)$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$



Power Terminology - 3

Several quantities are used so often in power calculations that they are given specific names. The third definition, the **Reactive Factor**, is the sine of the phase angle between voltage sinusoid and the current sinusoid. It should be clear that while the power factor was the coefficient in the Real Power, the Reactive Factor is the coefficient for Reactive Power.

Reactive Factor $\equiv rf \equiv \sin \theta$.

Using the alternative notation, we would say

Reactive Factor $\equiv rf \equiv \sin(\theta_v - \theta_i)$.

Note that rf is used as a common abbreviation for reactive factor.

$$v(t) = V_m \cos(\omega t + \theta) \text{ and}$$

$$i(t) = I_m \cos(\omega t); \text{ then}$$

$$Z = \frac{V(\omega)}{I(\omega)} = \frac{V_m}{I_m} \angle \theta.$$

$$\text{Real Power} = p_{AVERAGE} = P = \frac{V_m I_m}{2} \cos(\theta)$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$

Power Terminology - 4

Several quantities are used so often in power calculations that they are given specific names. When we have an inductor, the phase of the impedance, θ , is positive and equal to 90° . When we have a combination of passive elements where the inductances are dominant, this phase will be positive, but typically not 90° . We call this situation an inductive circuit, or an **inductive load**.

An Inductive Load means that $\theta > 0$.

Note that in this case the $\sin(\theta)$ will be positive, so the reactive power Q that is absorbed will be positive.

$$v(t) = V_m \cos(\omega t + \theta) \text{ and } i(t) = I_m \cos(\omega t); \text{ then}$$

$$Z = \frac{V(\omega)}{I(\omega)} = \frac{V_m}{I_m} \angle \theta.$$

$$\text{Real Power} = p_{\text{AVERAGE}} = P = \frac{V_m I_m}{2} \cos(\theta)$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$

Power Terminology - 5

Several quantities are used so often in power calculations that they are given specific names. When we have a combination of passive elements where the inductances are dominant, this phase will be positive, but typically not 90° . This means that the **current lags the voltage**, that is, the current appears to be behind the voltage if they are plotted on the same axes. When this happens, when we have an inductive load, we say we have a **lagging power factor**.

An Inductive Load means that $\theta > 0$,
and that we have a Lagging Power Factor.

Note that in this case the $\sin(\theta)$ will be positive, so the reactive power Q that is absorbed will be positive. We say that Reactive Power is being absorbed.

$$v(t) = V_m \cos(\omega t + \theta) \text{ and } i(t) = I_m \cos(\omega t); \text{ then}$$

$$Z = \frac{V(\omega)}{I(\omega)} = \frac{V_m}{I_m} \angle \theta.$$

$$\text{Real Power} = p_{\text{AVERAGE}} = P = \frac{V_m I_m}{2} \cos(\theta)$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$



Power Terminology - 6

Several quantities are used so often in power calculations that they are given specific names. When we have an capacitor, the phase of the impedance, θ , is negative, and equal to -90° . When we have a combination of passive elements where the capacitances are dominant, this phase will be negative, but typically not -90° . We call this situation a capacitive circuit, or a **capacitive load**.

A Capacitive Load means that $\theta < 0$.

Note that in this case the $\sin(\theta)$ will be negative, so the reactive power Q that is absorbed will be negative.

$$v(t) = V_m \cos(\omega t + \theta) \text{ and } i(t) = I_m \cos(\omega t); \text{ then}$$

$$Z = \frac{V(\omega)}{I(\omega)} = \frac{V_m}{I_m} \angle \theta.$$

$$\text{Real Power} = p_{\text{AVERAGE}} = P = \frac{V_m I_m}{2} \cos(\theta)$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$

Power Terminology - 7

Several quantities are used so often in power calculations that they are given specific names. When we have a combination of passive elements where the capacitances are dominant, this phase will be negative, but typically not -90° . This means that the **current leads the voltage**, that is, the current appears to be ahead of the voltage if they are plotted on the same axes. When this happens, when we have a capacitive load, we say we have a **leading power factor**.

A Capacitive Load means that $\theta < 0$,
and that we have a Leading Power Factor.

Note that in this case the $\sin(\theta)$ will be negative, so the reactive power Q that is absorbed will be negative. We say that Reactive Power is being delivered.

$$v(t) = V_m \cos(\omega t + \theta) \text{ and } i(t) = I_m \cos(\omega t); \text{ then}$$
$$Z = \frac{V(\omega)}{I(\omega)} = \frac{V_m}{I_m} \angle \theta.$$

$$\text{Real Power} = p_{\text{AVERAGE}} = P = \frac{V_m I_m}{2} \cos(\theta)$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$



Usefulness of Reactive Power

When we have inductive loads, such as motors, connected by long power lines, there is the potential for energy loss. In this case, we have energy that is being transmitted through the line to the load, only to be returned back through the line, from the load to the source. This causes energy to be lost. If the load can be adjusted to appear like a resistor, then this energy does not need to flow back and forth through the transmission lines, reducing the losses.

The solution is to connect capacitors near the motors. This makes the loads look like they are resistors. What happens is that the energy needed by the inductors are provided by the capacitors, moving back and forth between them. Thus, this energy only needs to travel through the transmission line once.

Reactive Power is a way to keep track of this phenomenon. By minimizing the Reactive Power, we can reduce losses.

$$\text{Real Power} = p_{\text{AVERAGE}} = P = \frac{V_m I_m}{2} \cos(\theta)$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$

Absorbing and Delivering Reactive Power

We are familiar with the idea of a resistor, which absorbs positive power. That is, the voltage times the current, in the passive sign convention, gives the power absorbed by the resistor, which will be positive.

Using this concept, we say that when we use the passive sign relationship, if the Reactive Power we solve for is positive, we will say that Reactive Power is being absorbed. Similarly, if Reactive Power is negative, we will say that Reactive Power is being delivered. We can show that, because of the phases, inductors absorb positive Reactive Power, and capacitors deliver positive Reactive Power.

$$\text{Real Power} = p_{\text{AVERAGE}} = P = \frac{V_m I_m}{2} \cos(\theta)$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$

Absorbing and Delivering Reactive Power – Note

We can show that, because of the phases, inductors absorb positive Reactive Power, and capacitors deliver positive Reactive Power.

It is very important to remember that, in fact, inductors and capacitors do not deliver or absorb power, on average. They take energy in, store it, and then return it. The Reactive Power is a measure of how energy is stored temporarily in sinusoidal systems, and the sign indicates whether it was stored in electric fields or magnetic fields.

$$\text{Real Power} = p_{\text{AVERAGE}} = P = \frac{V_m I_m}{2} \cos(\theta)$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$



So what is the point of all this?

- This is a good question. First, our premise is that since electric power is usually distributed as sinusoids, the issue of sinusoidal power is important.
- The quantities real and reactive power, that we have described here, are very different. Real power is the average power, and has direct meaning. Reactive power is a measure of power that is being stored temporarily. The sign tells us of the nature of the storage. Using these concepts, we can make changes which can improve the efficiency of the transmission of power.
- All of this is made even more useful, when we see how phasors can make the calculation of real and reactive power easier.





Complex Power

We have defined real power and reactive power, and indicated why they might be important for efficient power distribution when we have sinusoidal voltages and currents.

Now we are going to show how they can be found more easily. We will find that:

- A new concept called **complex power** can be defined in terms of complex numbers, as a function of the real power and reactive power.
- Using phasor analysis makes it relatively simple to find this complex power.
- This complex power tells us the real and reactive power, things that we want to know.

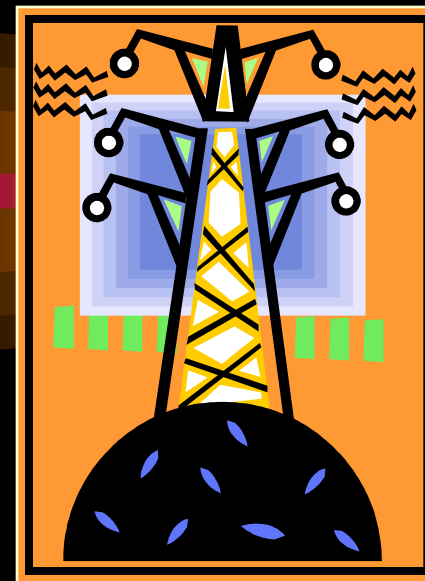


The power lines, which connect us from distant power generating systems, result in lost power. However, this lost power can be reduced by adjustments in the loads. This led to the use of the concept of reactive power.



Definition of Complex Power

We define **Complex Power** to be the **Real Power** added to **Reactive Power** times j , which is the square root of minus one. Thus, complex power is a complex number. We use the capital letter S to refer to complex power. The real power is the real part of the complex power. The reactive power is the imaginary part of the complex power.



Given that:

$$v(t) = V_m \cos(\omega t + \theta) \text{ and}$$

$$i(t) = I_m \cos(\omega t); \text{ then}$$

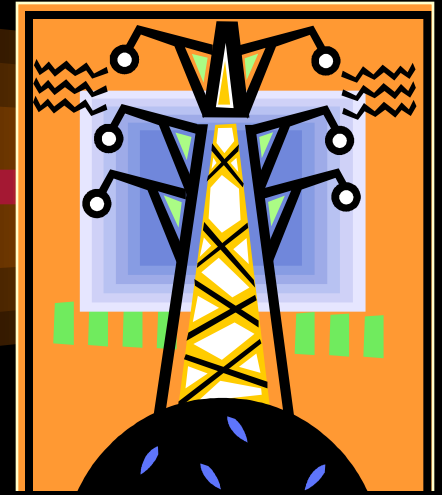
$$\text{Real Power} = P = \frac{V_m I_m}{2} \cos(\theta). \quad \text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$

$$\text{Complex Power} = S = P + jQ.$$



Definition of Apparent Power

We define **Apparent Power** to be the magnitude of the Complex Power. Thus, apparent power is a real number. We use brackets around the capital letter, $|S|$ to refer to apparent power. The apparent power is the magnitude of the complex power, and has the same units as complex power.



Given that:

$$v(t) = V_m \cos(\omega t + \theta) \text{ and} \\ i(t) = I_m \cos(\omega t); \text{ then}$$

$$\text{Real Power} = P = \frac{V_m I_m}{2} \cos(\theta).$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$

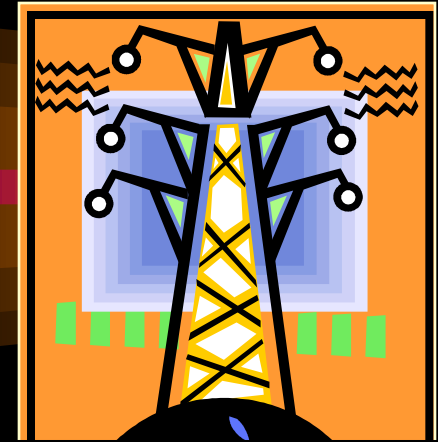
$$\text{Complex Power} = S = P + jQ.$$

$$\text{Apparent Power} = |S| = |P + jQ|.$$



Units for Complex Power

We use special units to keep all this straight. For **Complex Power** and **Apparent Power** we use the units [Volt-Amperes] or [VA]. For **Real Power** we use the units [Watts] or [W]. For **Reactive Power** we use the units [Volt-Amperes-Reactive] or [VAR]. It is important to use the correct units, so that we know what kind of power we are talking about.



Given that:

$$v(t) = V_m \cos(\omega t + \theta) \text{ and } i(t) = I_m \cos(\omega t); \text{ then}$$

Units are [W]

$$\text{Real Power} = P = \frac{V_m I_m}{2} \cos(\theta).$$

Units are [VAR]

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$

Units are [VA]

$$\text{Complex Power} = S = P + jQ.$$

$$\text{Apparent Power} = |S| = |P + jQ|.$$

Meaning of Complex Power

The meaning of complex power is much less direct than the definitions of real power and reactive power. One way to look at it is that complex power is a way to obtain the real power and reactive power, quickly and efficiently, using phasors. We will explain how, next.

Given that:

$$v(t) = V_m \cos(\omega t + \theta) \text{ and} \\ i(t) = I_m \cos(\omega t); \text{ then}$$

$$\text{Real Power} = P = \frac{V_m I_m}{2} \cos(\theta). \quad \text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta).$$

$$\text{Complex Power} = S = P + jQ.$$

The Usefulness of Complex Power – Part 1

We will show that it is relatively easy to obtain the complex power, if we use phasor analysis. Notice that we can substitute into our definition of complex power, using the formulas for real power and reactive power. We do this in the equations that follow. For this derivation, it is convenient to go back to our alternative notation, which we called the General Case, where the phase of the voltage is θ_v , and the phase of the current is θ_i .

$$\text{Real Power} = P = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i). \quad \text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta_v - \theta_i).$$

Given that:

$$v(t) = V_m \cos(\omega t + \theta_v) \text{ and} \\ i(t) = I_m \cos(\omega t + \theta_i); \text{ then}$$

$$\text{Complex Power} = S = P + jQ =$$

$$S = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) + j \frac{V_m I_m}{2} \sin(\theta_v - \theta_i) =$$

$$S = \frac{V_m I_m}{2} (\cos(\theta_v - \theta_i) + j \sin(\theta_v - \theta_i)).$$

The Usefulness of Complex Power – Part 2

Using our result from the previous slide, we may recognize that this is in the form of Euler's Relation. Euler's Relation is given below to remind us of what it says. Thus, we can express complex power in terms of a complex exponential. We have the equations that follow.

$$\text{Real Power} = P = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i).$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta_v - \theta_i).$$

$$\text{Complex Power} = S = P + jQ =$$

$$S = \frac{V_m I_m}{2} (\cos(\theta_v - \theta_i) + j \sin(\theta_v - \theta_i)) =$$

$$S = \frac{V_m I_m}{2} e^{j(\theta_v - \theta_i)} = \frac{V_m I_m}{2} e^{j(\theta_v)} e^{j(-\theta_i)}.$$

Given that:

$$v(t) = V_m \cos(\omega t + \theta_v) \text{ and } i(t) = I_m \cos(\omega t + \theta_i); \text{ then}$$

Euler's Relation says that

$$e^{j\phi} = \cos(\phi) + j \sin(\phi).$$

The Usefulness of Complex Power – Part 3

Now, we take this equation from the previous slide, and combine the terms for voltage together, and the terms for current together, and we recognize the result as the phasor transforms of the voltage and current ($V_m(\omega)$ and $I_m(\omega)$).

$$\text{Real Power} = P = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i).$$

Given that:

$$v(t) = V_m \cos(\omega t + \theta_v) \text{ and } i(t) = I_m \cos(\omega t + \theta_i); \text{ then}$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta_v - \theta_i).$$

$$\text{Complex Power} = S = P + jQ =$$

$$S = \frac{V_m I_m}{2} e^{j(\theta_v - \theta_i)} = \frac{V_m I_m}{2} e^{j(\theta_v)} e^{j(-\theta_i)} =$$

$$S = \frac{V_m e^{j(\theta_v)} I_m e^{j(-\theta_i)}}{2} = \frac{V_m(\omega) I_m^*(\omega)}{2}.$$

The Usefulness of Complex Power – Part 4

Note that the phasor transform of the current $I(\omega)$ would have a phase θ_i . Since the phase here is the negative of that, $-\theta_i$, we don't get the phasor transform, but rather the complex conjugate of the phasor transform, or $I^*(\omega)$.

$$\text{Real Power} = P = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i).$$

$$\text{Reactive Power} = Q = \frac{V_m I_m}{2} \sin(\theta_v - \theta_i).$$

Given that:

$$v(t) = V_m \cos(\omega t + \theta_v) \text{ and } i(t) = I_m \cos(\omega t + \theta_i); \text{ then}$$

$$\text{Complex Power} = S = P + jQ =$$

$$S = \frac{V_m I_m}{2} e^{j(\theta_v - \theta_i)} = \frac{V_m I_m}{2} e^{j(\theta_v)} e^{j(-\theta_i)} =$$

$$S = \frac{V_m e^{j(\theta_v)} I_m e^{j(-\theta_i)}}{2} = \frac{V_m(\omega) I_m^*(\omega)}{2}.$$

The Usefulness of Complex Power – Part 5

Thus, we reach the conclusion about complex power, which is shown below. We get the complex power by multiplying the phasor of the voltage times the complex conjugate of the phasor for the current. The real part of this is the Real Power, and the imaginary part of this is the reactive power.

$$\text{Complex Power} = S = P + jQ = \frac{V_m(\omega)I_m^*(\omega)}{2}.$$

If we use a different way of defining phasors, where the magnitude of the phasor is the RMS value of the sinusoid, instead of the zero-to-peak value, we then have

$$\text{Complex Power} = S = P + jQ = V_{rms}(\omega)I_{rms}^*(\omega).$$



The Usefulness of Complex Power – Part 6

This, then is the fundamental usefulness of complex power. We want to know P and Q . The real power, P , is the average power. The reactive power, Q , is a measure of the energy delivered to inductors or capacitors, and then returned. We can get these by taking the product of phasor voltage and the complex conjugate of the phasor current, using rms phasors. The real part of this product is P , and the imaginary part is Q .

$$\text{Complex Power} = S = P + jQ = V_{rms}(\omega) I_{rms}^*(\omega).$$

The Usefulness of Complex Power – Part 7

There is an approach, based on this, that can be used to find P and Q , which is even easier to use. Using the notation for impedance,

$$\text{Impedance} = Z = R + jX = \frac{V_{rms}}{I_{rms}}$$

where R is called the resistance, and X is called the reactance, we can then say that

$$\begin{aligned} S &= P + jQ = V_{rms}(\omega) I_{rms}^*(\omega) = \\ P + jQ &= (I_{rms}(\omega) Z) I_{rms}^*(\omega) = |I_{rms}(\omega)|^2 Z = \\ P + jQ &= |I_{rms}(\omega)|^2 (R + jX). \end{aligned}$$

The Usefulness of Complex Power – Part 8

Thus, to find P and Q , we can use

$$P = |I_{rms}(\omega)|^2 (R), \text{ and}$$

$$Q = |I_{rms}(\omega)|^2 (X).$$

Note that we don't need the phasor for the voltage, or even the phase of the phasor for the current. All we need is the magnitude of the phasor for the current, and the impedance, where

$$\text{Impedance} = Z = R + jX.$$

Notation with RMS Phasors

The equations that follow,

$$S = P + jQ = V_{rms}(\omega) I_{rms}^*(\omega) =$$

$$P + jQ = (I_{rms}(\omega) Z) I_{rms}^*(\omega) = |I_{rms}(\omega)|^2 Z =$$

$$P + jQ = |I_{rms}(\omega)|^2 (R + jX).$$

use rms phasors. Here, we have added rms to the subscript. In our handwritten notation, we could also simply omit the *m* from the subscript. **Thus, we can assume that if there is no *m* added to the subscript of a phasor, it is an rms phasor.**

Important Notes – 1

When we find P and Q , we need to be clear about what our answer means. In these derivations, we have assumed that we were using voltage and current in the passive sign relationship. However, this is not always true.

$$P_{abs,load} = \left| \mathbf{I}_{load,rms}(\omega) \right|^2 (R_{load}), \text{ and}$$
$$Q_{abs,load} = \left| \mathbf{I}_{load,rms}(\omega) \right|^2 (X_{load}).$$

So, we need to return to our practice of having a two-part subscript for every power expression. This includes real power, reactive power, and complex power.

Important Notes – 2

When we find P and Q , we need to be clear about what our answer means. So, our rules should be

$$S_{abs,thing} = P_{abs,thing} + jQ_{abs,thing} = V_{x,rms}(\omega)I_{x,rms}^*(\omega),$$

where $V_{x,rms}(\omega)$ and $I_{x,rms}(\omega)$ are defined in the passive sign relationship.

Signs matter, as always. If we want to get power absorbed, as shown here, we will often use the passive sign relationship. If we use the active sign relationship, we need to include a minus sign.

Important Notes – 3

When we find P and Q , we need to be clear about what our answer means.

$$S_{abs,thing} = P_{abs,thing} + jQ_{abs,thing} = -V_{x,rms}(\omega)I_{x,rms}^*(\omega),$$

where $V_{x,rms}(\omega)$ and $I_{x,rms}(\omega)$ are defined in the active sign relationship.

Signs matter, as always. If we want to get power absorbed, as shown here, we will often use the passive sign relationship. If we use the active sign relationship, we need to include a minus sign.

Important Notes – 4

There is a useful concept here. If you look at these equations, and compare them with the definitions for these quantities, we have

$$P_{abs.by.thing} = |I_{rms}(\omega)|^2 (R), \text{ and}$$

$$Q_{abs.by.thing} = |I_{rms}(\omega)|^2 (X), \text{ where}$$

$$Z = R + jX \text{ and } S = P + jQ.$$

This means that the phase of the impedance of some load, is equal to the phase of the complex power for that load. The phase of the complex power for the load is called the **power factor angle**.

Important Notes – 5

The phase of the impedance of a load is equal to the phase of the complex power for that load, and is called the **power factor angle**.

$$P_{abs.by.thing} = |I_{rms}(\omega)|^2 (R), \text{ and}$$

$$Q_{abs.by.thing} = |I_{rms}(\omega)|^2 (X), \text{ where}$$

$$Z = R + jX \text{ and } S = P + jQ.$$

This is useful because for a load, we often want the reactive power to be zero. This corresponds to a zero power factor angle. Thus, the power factor angle is a useful quantity to measure and know, and it is equal to phase of the impedance of the load.

Important Notes – 6

The complex power absorbed by a load can also be expressed in terms of the phasor voltage across that load.

$$S_{abs,load} = \frac{|V_{rms}(\omega)|^2}{Z_{Load}^*} = \frac{|V_{rms}(\omega)|^2}{R_{Load} - jX_{Load}}. \text{ However,}$$

~~$$P_{abs,load} \neq \frac{|V_{rms}(\omega)|^2}{R_{Load}} \text{ and } Q_{abs,load} \neq \frac{|V_{rms}(\omega)|^2}{-X_{Load}}.$$~~

This is not as useful as the formulas for the current, so I suggest that you do not use this approach.



Sample Problem 1

In the circuit shown in Fig. 10.13, a load having an impedance of $39 + j26 \Omega$ is fed from a voltage source through a line having an impedance of $1 + j4 \Omega$. The effective, or rms, value of the source voltage is 250 V.

- Calculate the load current I_L and voltage V_L .
- Calculate the average and reactive power delivered to the load.
- Calculate the average and reactive power delivered to the line.
- Calculate the average and reactive power supplied by the source.

Solutions:

$$P_{\text{abs.by.load}} = 975[\text{W}]$$

$$Q_{\text{abs.by.load}} = 650[\text{VAR}]$$

$$P_{\text{abs.by.line}} = 25[\text{W}]$$

$$Q_{\text{abs.by.line}} = 100[\text{VAR}]$$

Taken from
Nilsson and
Riedel, 8th
Edition

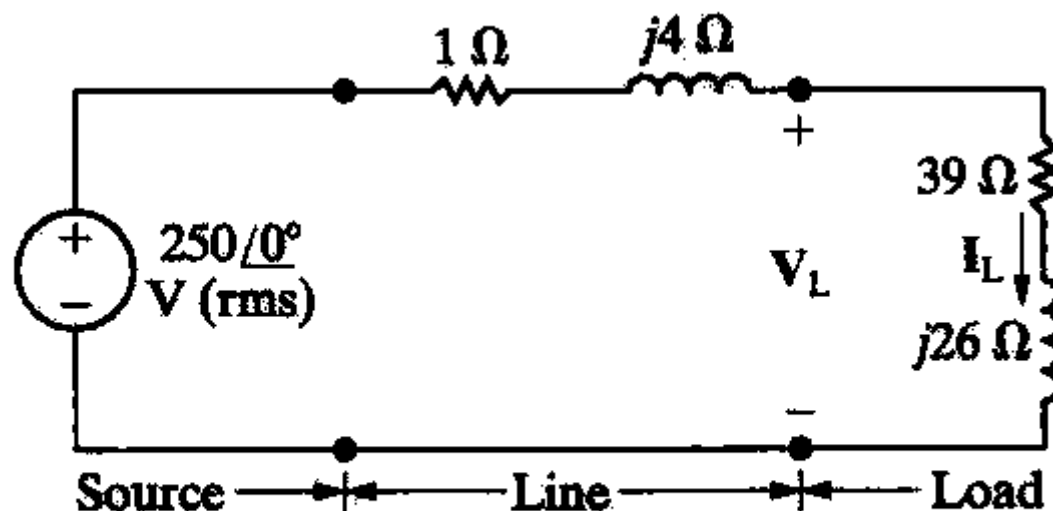


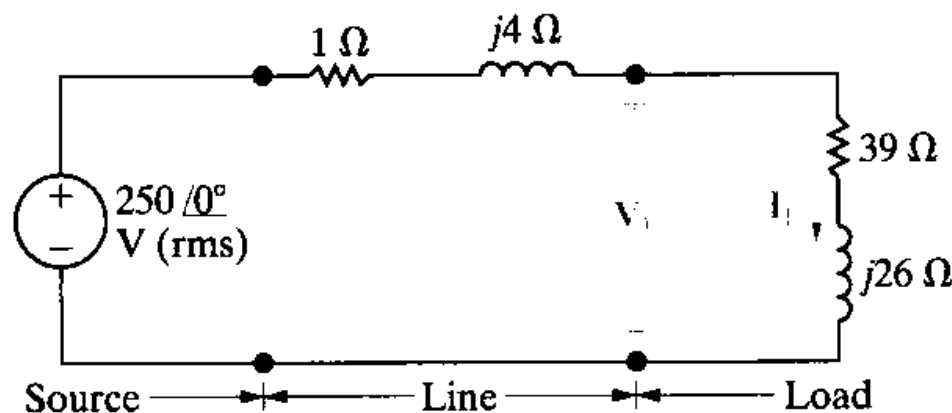
Figure 10.13 ▲ The circuit for Example 10.5.



Sample Problem 2

10.4 The load impedance in the circuit shown is shunted by a capacitor having a capacitive reactance of -52Ω . Calculate:

- the rms phasors V_L and I_L ,
- the average power and magnetizing reactive power absorbed by the $(39 + j26) \Omega$ load impedance,
- the average power and magnetizing reactive power absorbed by the $(1 + j4) \Omega$ line impedance,
- the average power and magnetizing reactive power delivered by the source, and
- the magnetizing reactive power delivered by the shunting capacitor.



Taken from
Nilsson and
Riedel, 8th
Edition

Solutions:

$$P_{\text{abs.by.load}} = 1129[\text{W}]$$

$$Q_{\text{abs.by.load}} = 753[\text{VAR}]$$

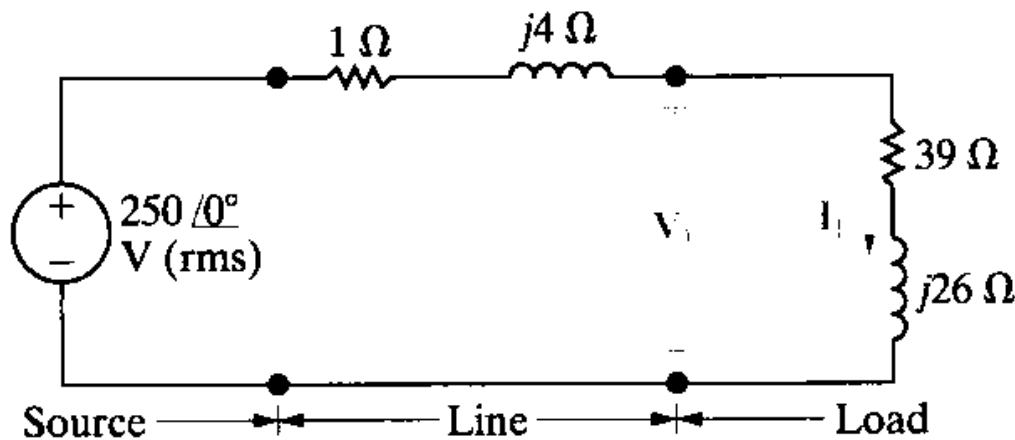
$$P_{\text{abs.by.line}} = 23.52[\text{W}]$$

$$Q_{\text{abs.by.line}} = 94.09[\text{VAR}]$$



Sample Problem 3

For the circuit shown below, assume that a capacitor with a reactance equal to X_C is placed in parallel with the load. Find the value for that reactance that will give a combined load with a unity power factor. Using that reactance value, find the real and reactive power absorbed by the load, and the real and reactive power absorbed by the line.



Solutions:

$$X_C = -84.5[\Omega]$$

$$P_{\text{abs.by.load}} = 1066[\text{W}]$$

$$Q_{\text{abs.by.load}} = 0$$

$$P_{\text{abs.by.line}} = 18.92[\text{W}]$$

$$Q_{\text{abs.by.line}} = 75.68[\text{VAR}]$$



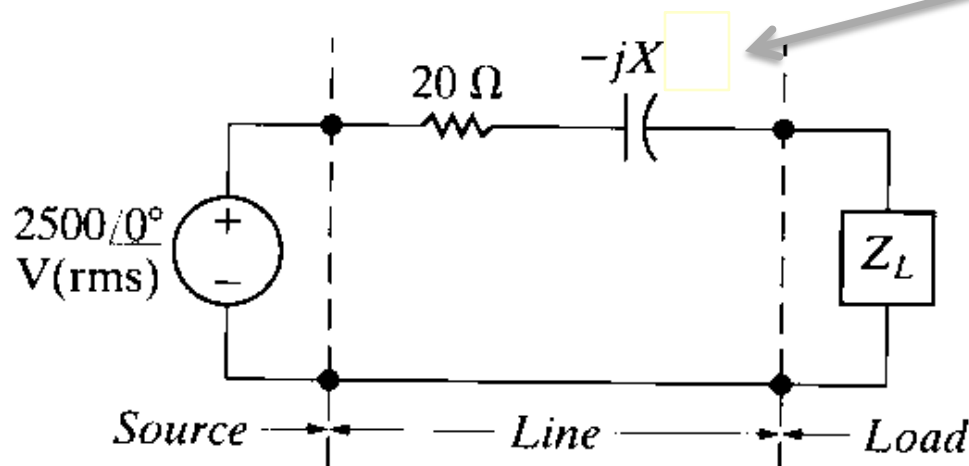
Sample Problem 4

10.7 The load impedance in Fig. P10.7 absorbs 40 kW and 30 [redacted] kvars. The sinusoidal voltage source develops 50 kW.

- Find the values of capacitive line reactance that will satisfy these constraints.
- For each value of line reactance found in (a), show that the [redacted] developed equals the [redacted] absorbed.

Taken from Nilsson and Riedel, 8th Edition, with some edits.

Figure P10.7



We do not show units for variables. So, we will remove these inappropriate units.



Sample Problem 5

The circuit given below operates in steady-state.

We know that Load 1 absorbs 37[kW] and delivers 56[kVAR].

We know that Load 2 absorbs $(43 \angle -16^\circ)$ [kVA].

We know that Load 3 absorbs 57[kW] at a lagging power factor of 0.47.

a) Find $v_A(t)$.

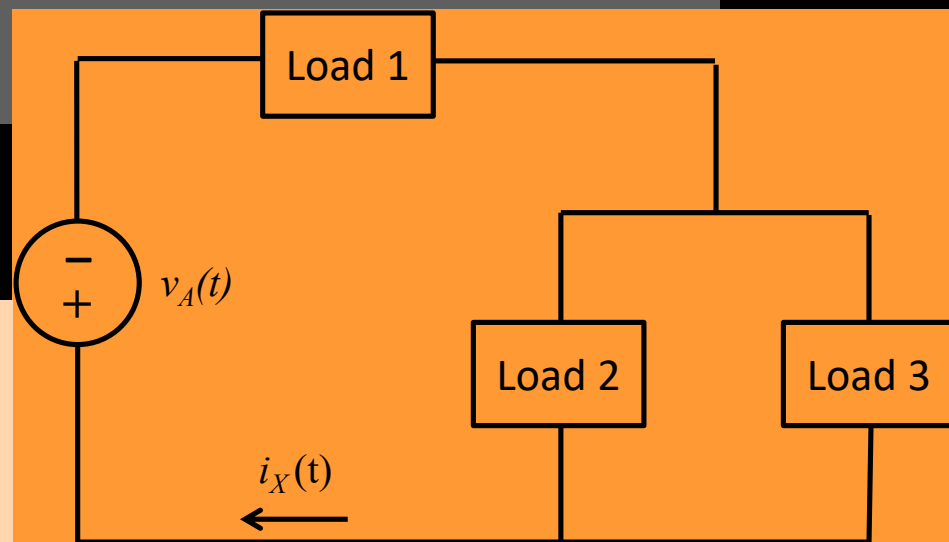
b) Find the impedance of Load 1.

Taken from Final Exam,
Fall 2015, Problem 7

Solutions:

a) $v_A(t) = 5,338[\text{V}] \cos \left(350 \left[\frac{\text{rad}}{\text{s}} \right] t + 63.09^\circ \right)$

b) $(26.59 - 40.25j)[\Omega]$



$$i_X(t) = 52.7 \sin \left(350 \left[\frac{\text{rad}}{\text{s}} \right] t - 43^\circ \right) [\text{A}]$$



So what is the point of all this?

- This is a good question. First, our premise is that since electric power is usually distributed as sinusoids, the issue of sinusoidal power is important.
- Second, the quantities real and reactive power are important. Real power is the average power, and has direct meaning. Reactive power is a measure of the energy that is being stored temporarily. The sign tells us of the nature of the storage. Using these concepts, we can make changes which can improve the efficiency of the transmission of power.
- Phasors make the calculation of real and reactive power straightforward. We use the new quantity “complex power” to tie it all together. The complex power gives us real and reactive power relatively easily.





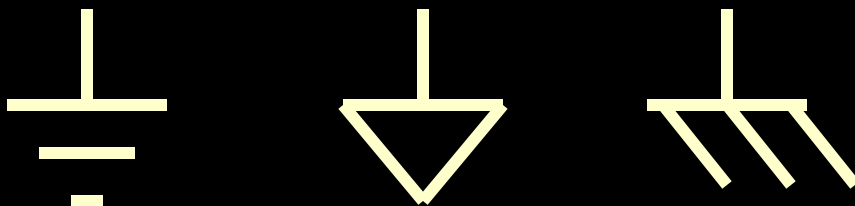
Notation

- We make a big distinction between things that vary (signals) and things that stay the same (DC power supplies). We will use a shorthand notation to make these distinctions easy to convey. In fact, we use a variety of commonly accepted conventions as we move forward. A set of conventions that we will use follows.



Notation

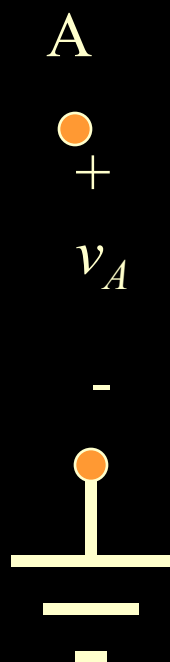
- The reference point for voltages is usually defined, and called **ground**, or **common**. Ground is the more common term, although it may have no relationship to the potential of the earth.
- Below we show some common symbols for common or ground.





Notation

- v_A , V_A , v_a , V_a – all of these refer to the voltage at point A with respect to ground. Notice that there is a polarity defined by this notation. This notation also means that we do not have to label the + and – signs on a circuit schematic to define the voltage. Once point A is labeled, and ground is shown or known, the voltages v_A , V_A , v_a , and V_a , are defined.





Notation

- v_{AB} , V_{AB} , v_{ab} , V_{ab} - refer to the voltage at point A with respect to point B. Notice that there is a polarity defined by this. This notation also means that we do not have to label the + and – signs on a circuit schematic to define the voltage. Once points A and B are labeled, the voltages v_{AB} , V_{AB} , v_{ab} , and V_{ab} , are defined.

A



v_{AB}

-

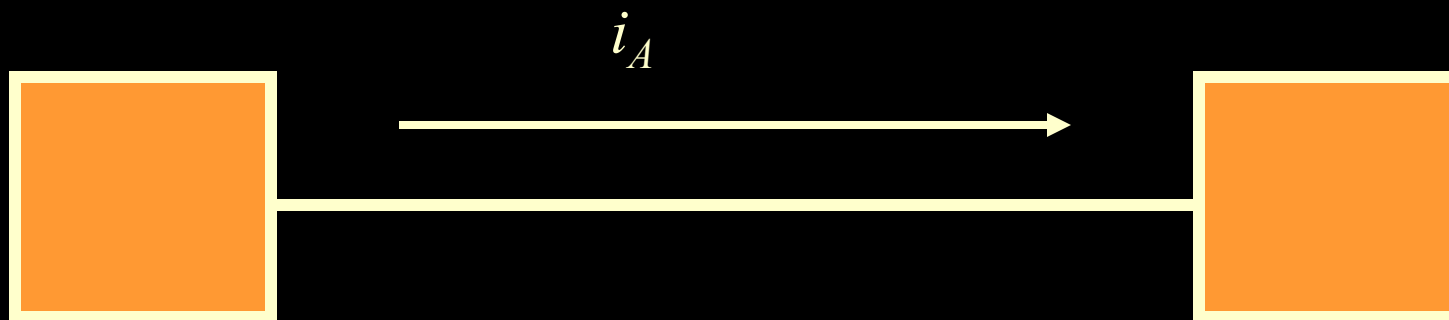


B



Notation

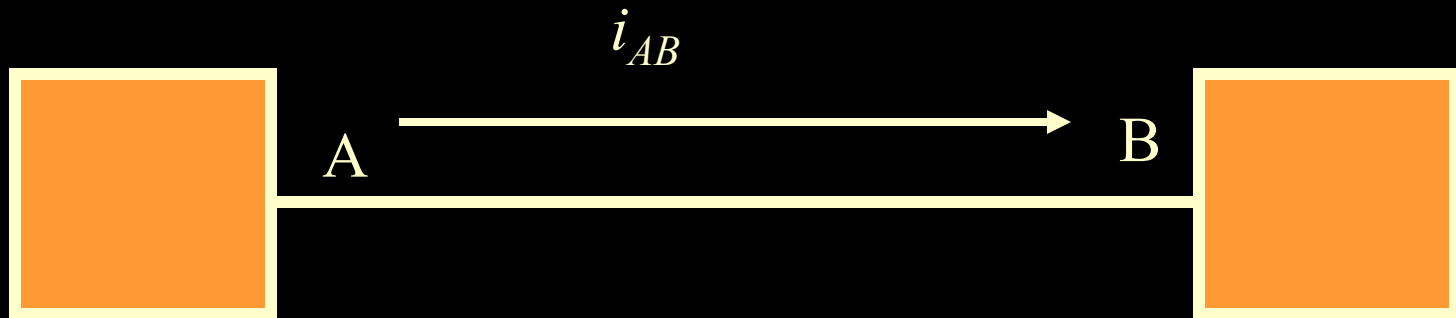
- Current polarities are shown with an arrow. Thus, current polarities must be defined, and the easiest way to do this is with an arrow on the circuit schematic.





Notation

- Current polarities are shown with an arrow. Thus, current polarities must be defined. Another way to do this is with ordered subscripts, with defined terminal names. The current i_{AB} flows from terminal A to terminal B.



Notation

- v_A is the total instantaneous quantity (lowercase_{UPPERCASE}).
- V_A is the dc component, nonvarying part of a quantity (UPPERCASE_{UPPERCASE}).
- v_a is the ac component, varying part of a quantity (lowercase_{lowercase}).
- The total instantaneous quantity is equal to the sum of the dc component and the ac component. That is, it is true that $v_A = V_A + v_a$.





Notation

- v_A is the total instantaneous quantity (lowercase_{UPPERCASE}).
- V_A is the dc component, nonvarying part of a quantity (UPPERCASE_{UPPERCASE}).
- v_a is the ac component, varying part of a quantity (lowercase_{lowercase}).
- **BACKGROUND:** Any quantity as a function of time can be broken down to a sum of a dc component (the average value or the mean value) and an ac component (a time-varying signal with zero mean).



Notation

- V_a is the phasor quantity (UPPERCASE_{lowercase}). In the textbook, Nilsson and Riedel use bold-face variables for phasors. We will use the same approach in these notes. Nilsson and Riedel then use V_a (UPPERCASE_{lowercase} but with no bold-face) to refer to quantities such as the coefficient of a sinusoid, or to an rms quantity, or other special quantities.



Notation

- $V_{a,rms}$ refers to an rms phasor value. Sometimes the rms subscript is left out. We will need to be able to judge from the context whether a phasor is an rms phasor or a magnitude-based phasor.